PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Malik Muhammad Faheem Badar
Program/Batch	B084-Networkwalks
Date	11 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own Virtual Lab Environment (local network discovery)
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

The activities documented in this report were performed within the authorized scope of the assigned internship tasks and my own virtual lab environment. The website reconnaissance was conducted with permission, while network discovery was limited to my personal lab network. No exploitation or unauthorized access was attempted. The observations are intended for educational and security assessment purposes only.

2. Introduction

This report documents the Week 2 practical activities completed during my Cybersecurity Internship at Networkwalks. The work covers two areas: Footprinting and Reconnaissance of the assigned website, networkwalks.com, and host discovery within my own virtual lab network.

The activities involved collecting publicly available domain and website information, examining DNS records and HTTP responses, identifying WAF technology, and discovering active hosts using Zenmap.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Environment for reconnaissance
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local network to find live hosts, IPs and MAC addresses.
Linux Terminal (ifconfig)	Identify Local Network Configuration

4. Activities Performed

4.1 Footprinting & Reconnaissance

Footprinting was performed on networkwalks.com using six Kali Linux tools to collect publicly available information about the domain and its web infrastructure.

WHOIS provided domain registration details, including the registrar, registration dates, and name servers. WhatWeb identified the website technologies, including Apache, WordPress 7.1.4, and WP Download Manager 3.3.76.

Nslookup resolved networkwalks.com to IP address 192.232.216.135. The cURL command was used to inspect HTTP response headers, which returned an HTTP/2 200 response and showed the WordPress REST API endpoint.

Wafw00f identified ModSecurity (SpiderLabs) as the detected Web Application Firewall. DNSRecon returned eight DNS records, including SOA, NS, MX, A, TXT, and SRV records.

4.2 Network Scanning with Zenmap

Network discovery was performed within my own virtual lab environment. The ifconfig command identified the local interface eth0 with IP address 10.0.0.2 and subnet mask 255.255.255.0. Zenmap was then used with the Ping Scan option to discover active hosts in the local network. The scan covered 256 addresses and identified two live hosts:

- 10.0.0.1
- 10.0.0.2

The scan results also displayed the MAC addresses of both hosts. This activity was limited to host discovery; no port or service scan was performed.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified Apache, WordPress 7.1.4 and WP Download Manager 3.3.76.	Attackers may use exposed technology and version details to research known vulnerabilities and identify potential attack vectors.	● **Medium**
2	Server IP address identifiable	Nslookup resolved networkwalks.com to 192.232.216.135.	The IP address may help attackers identify the hosting infrastructure and perform further reconnaissance.	● **Low**
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	Exposed technical details may assist attackers in identifying application components and planning further enumeration.	● **Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Identifying the WAF technology may help attackers research its configuration and potential weaknesses.	● **Low**
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information may help attackers understand the domain infrastructure and identify publicly exposed services.	● **Medium**
6	Multiple live hosts visible on local network	Zenmap discovered two live hosts: 10.0.0.1 and 10.0.0.2.	Host discovery provides visibility into active devices and may support further internal network reconnaissance.	● **Medium**

Risk level key: ● Critical ● Medium ● Low

The identified findings represent potential information security risks based on the reconnaissance and network discovery activities. They are not confirmed vulnerabilities.

No exploitation or vulnerability validation was performed during these exercises. Therefore, further authorized security testing would be required to determine whether any of the observations could lead to an exploitable weakness.

6. Recommendations

1. **Review publicly exposed technology information**
   Organizations should regularly review publicly visible information about their web technologies, CMS platforms, and installed plugins to reduce unnecessary information exposure.
2. **Keep software and plugins updated**
   CMS platforms, plugins, and web server software should be updated regularly to address known security issues and reduce potential attack risks.
3. **Review HTTP response headers**
   HTTP response headers should be reviewed to identify unnecessary technical information and ensure that appropriate security headers are configured.
4. **Review DNS records regularly**
   DNS records should be periodically reviewed to ensure that only necessary records and services are publicly accessible.
5. **Properly configure and monitor the WAF**
   The Web Application Firewall should be properly configured, regularly updated, and monitored to help detect and block suspicious web requests.
6. **Perform regular internal network discovery**
   Organizations should periodically perform authorized network discovery to identify active hosts and maintain visibility of connected devices.
7. **Investigate unknown network devices**
   Any unfamiliar or unauthorized devices identified during network discovery should be investigated to verify their identity and purpose.
8. **Maintain updated network documentation**
   Network information, including IP addresses, MAC addresses, and connected devices, should be documented and updated regularly.

7. Conclusion

During Week 2 of my Cybersecurity Internship at Networkwalks, I completed practical activities related to Footprinting and Reconnaissance, followed by Network Scanning and Host Discovery. These exercises provided practical experience in gathering information about a target website and identifying active devices within a controlled network environment.

During the footprinting activity, I used WHOIS, WhatWeb, Nslookup, cURL, Wafw00f, and DNSRecon to examine domain registration details, identify web technologies, resolve IP addresses, inspect HTTP response headers, detect WAF technology, and enumerate DNS records. The results provided an overview of the publicly available information associated with networkwalks.com.

For the network discovery activity, I used Linux network configuration commands and Zenmap to examine my virtual lab network. The Ping Scan identified two active hosts within the scanned network range, providing an understanding of basic network discovery and host identification.

Overall, these practical exercises strengthened my understanding of reconnaissance techniques, information gathering, DNS enumeration, and network discovery. They also highlighted the importance of reviewing exposed information, maintaining updated systems, and performing security assessments within an authorized scope. No exploitation or vulnerability validation was performed during these activities.

8. Evidences Collected

1. WHOIS – Domain registration information.



```
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability.  VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
connect: Network is unreachable
```

2. WhatWeb – Website technology identification.

```
└─$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache],
 HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,
x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@netw
orkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1
], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&amp;pidnV
ar2=50511&amp;prtVar2=7&amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Net
workwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], Wor
dPress[7.1.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, G
oogle-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Downl
oad Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&amp;pidnVar2=50511&amp;prtVar2=7&amp;scvVar2=12,applicati
on/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-
policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
```

3. Nslookup – Domain IP resolution.

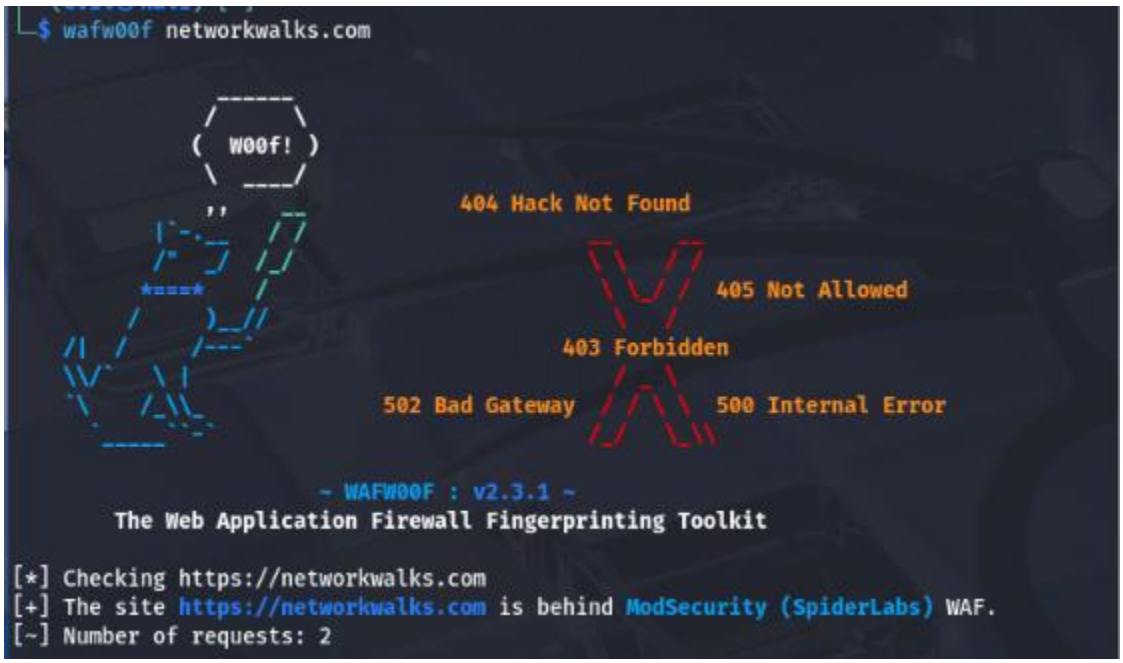
```
└─$ nslookup networkwalks.com
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

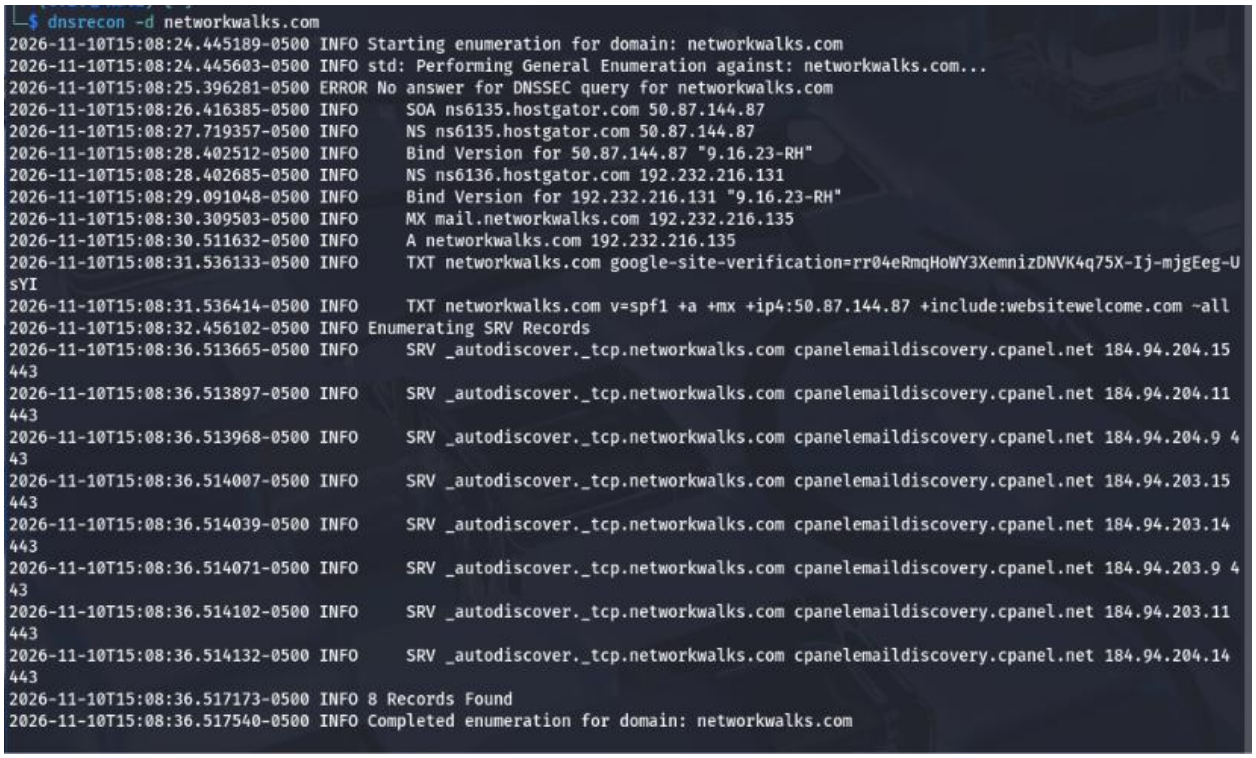
4. curl – HTTP response headers.

```
└─$ curl -I  https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.
net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "
https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="al
ternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=c1d4f29633197a2eea152958026713eb; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Tue, 10 Nov 2026 20:04:34 GMT
server: Apache
```

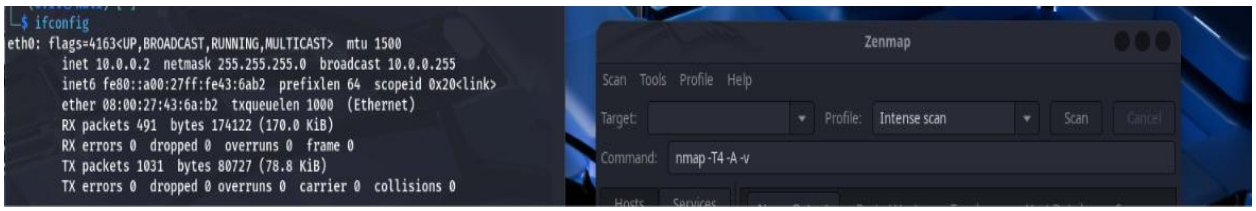
5. Wafw00f – WAF detection result.



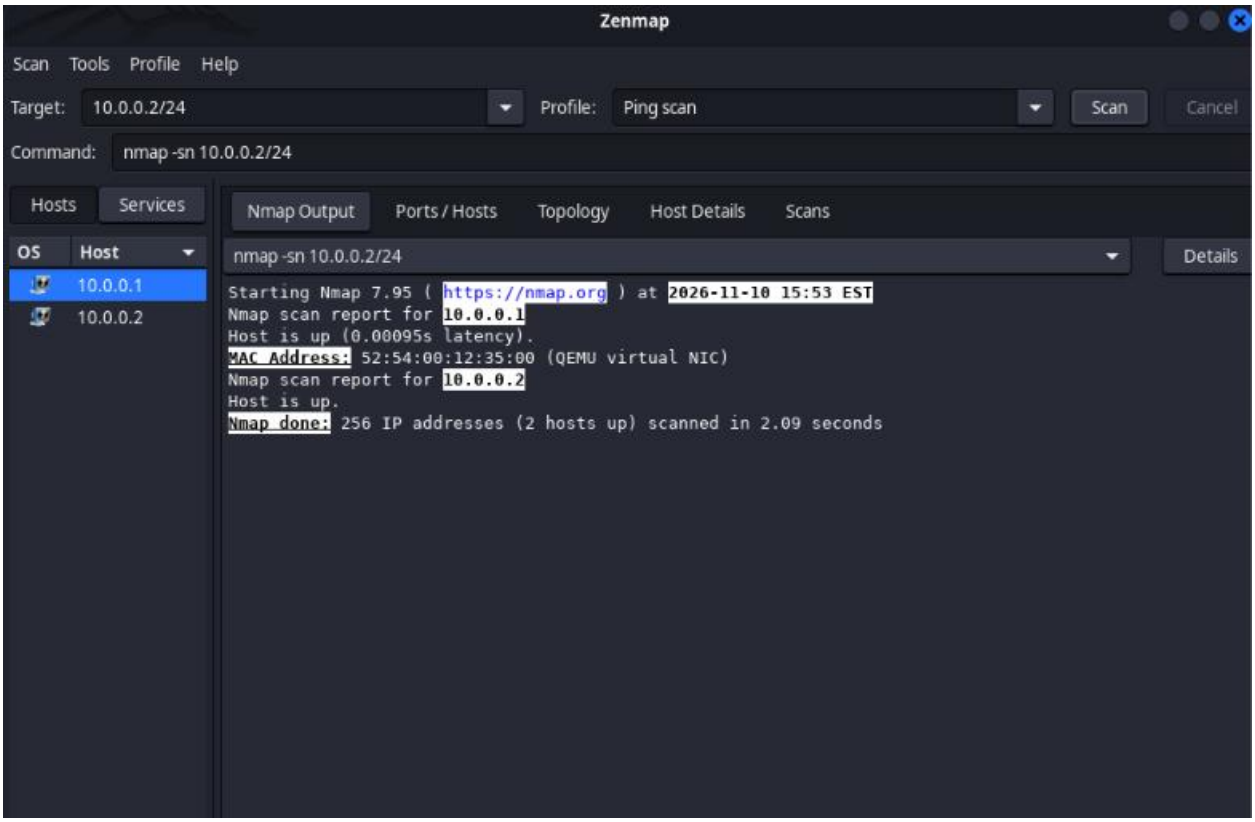
6. DNSRecon – DNS record enumeration.

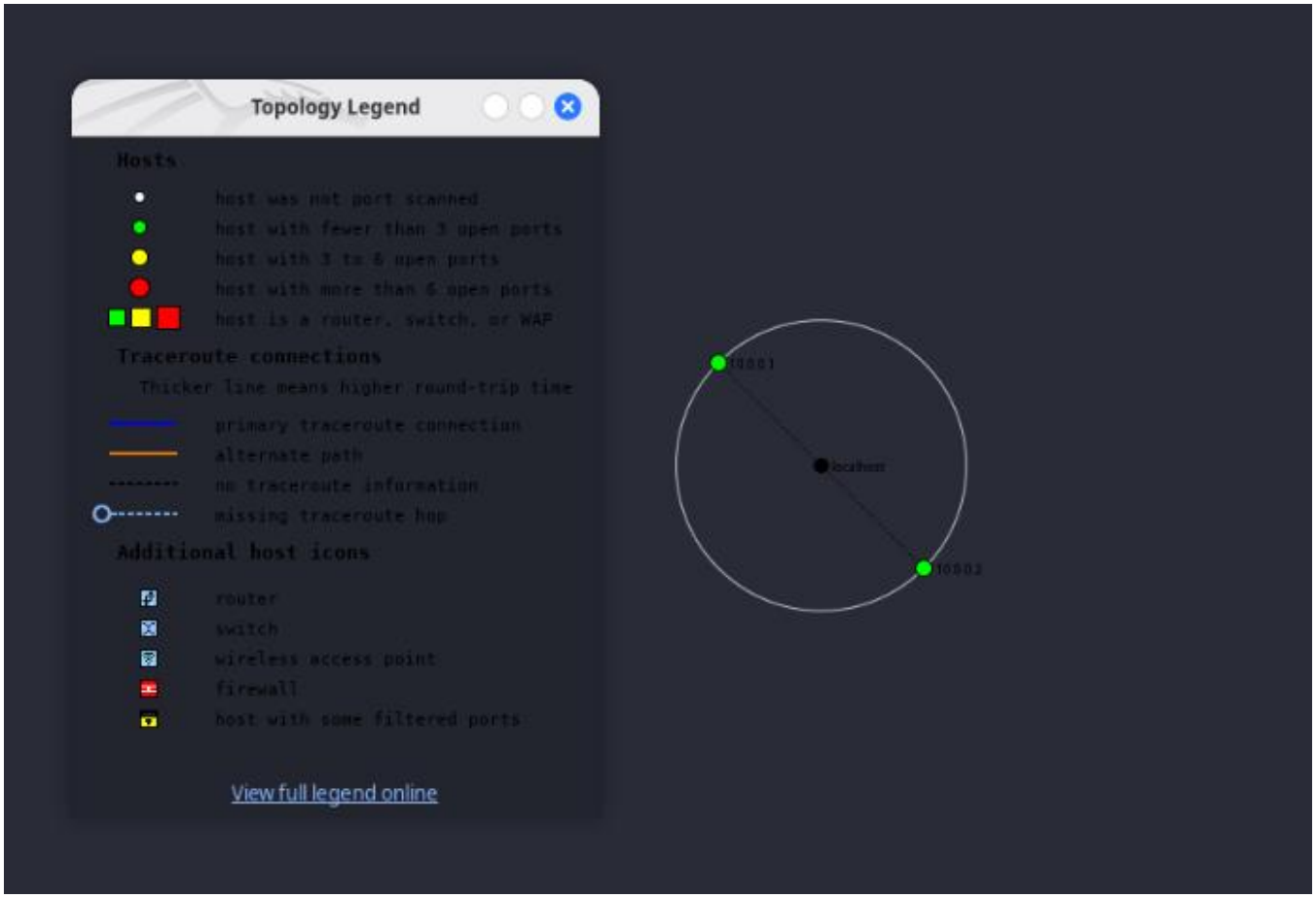


7. ifconfig – Local network configuration.



8. Zenmap – Ping Scan and active host discovery.





Author
Malik Muhammad Faheem Badar
Cybersecurity Intern
Batch : B084
LinkedIn: https://www.linkedin.com/in/malik-muhammad-faheem-badar-2b1a43414/

Project Information
Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub